

Experiment: 6

5G NR Frame Structure Analysis with Different Subcarrier Spacing

Objectives 1:

To analyze different Subcarrier Spacing (SCS) values used in 5G NR numerology

CODE:

```
clc;

clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

mu = 0:4; % Numerology Index

bar(scs)

grid on

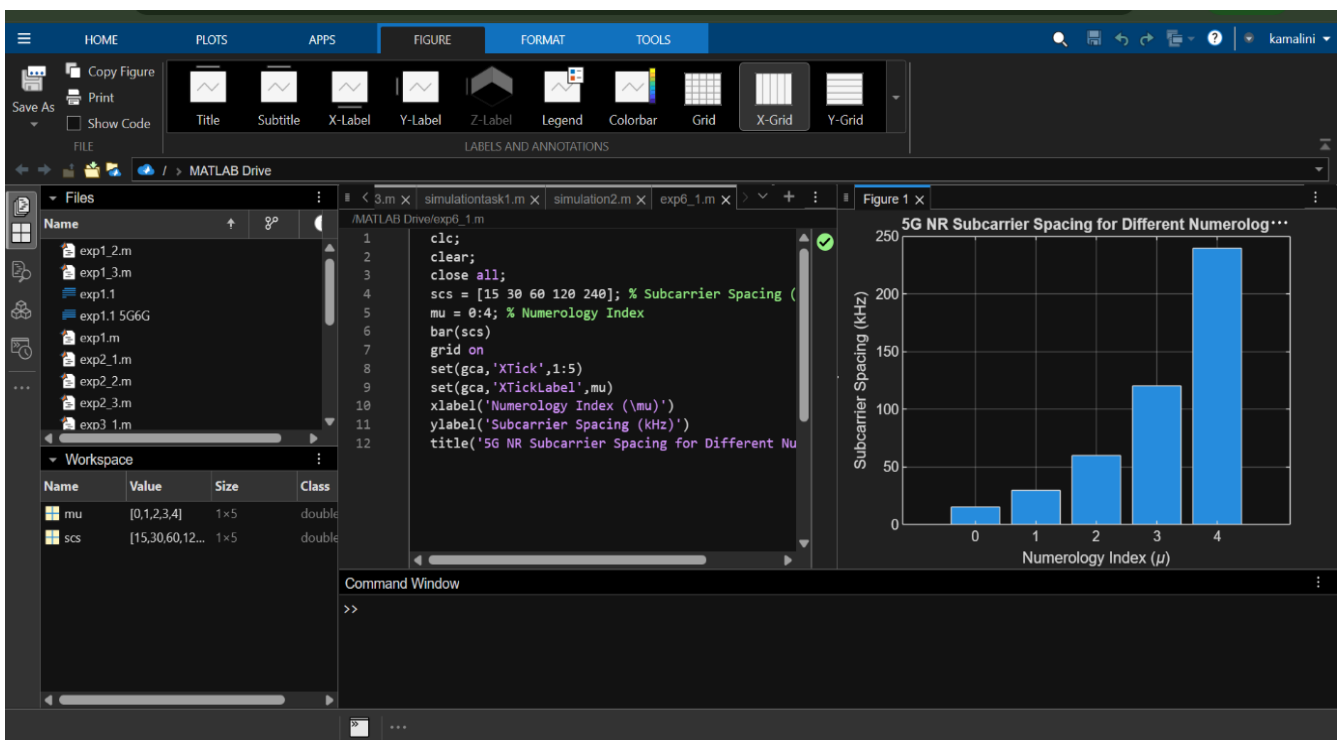
set(gca,'XTick',1:5)

set(gca,'XTickLabel',mu)

xlabel('Numerology Index (\mu)')

ylabel('Subcarrier Spacing (kHz)')

title('5G NR Subcarrier Spacing for Different Numerologies')
```



Objectives 2:

To compare the slot duration corresponding to different Subcarrier Spacing values in 5G NR.

CODE:

```
clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)

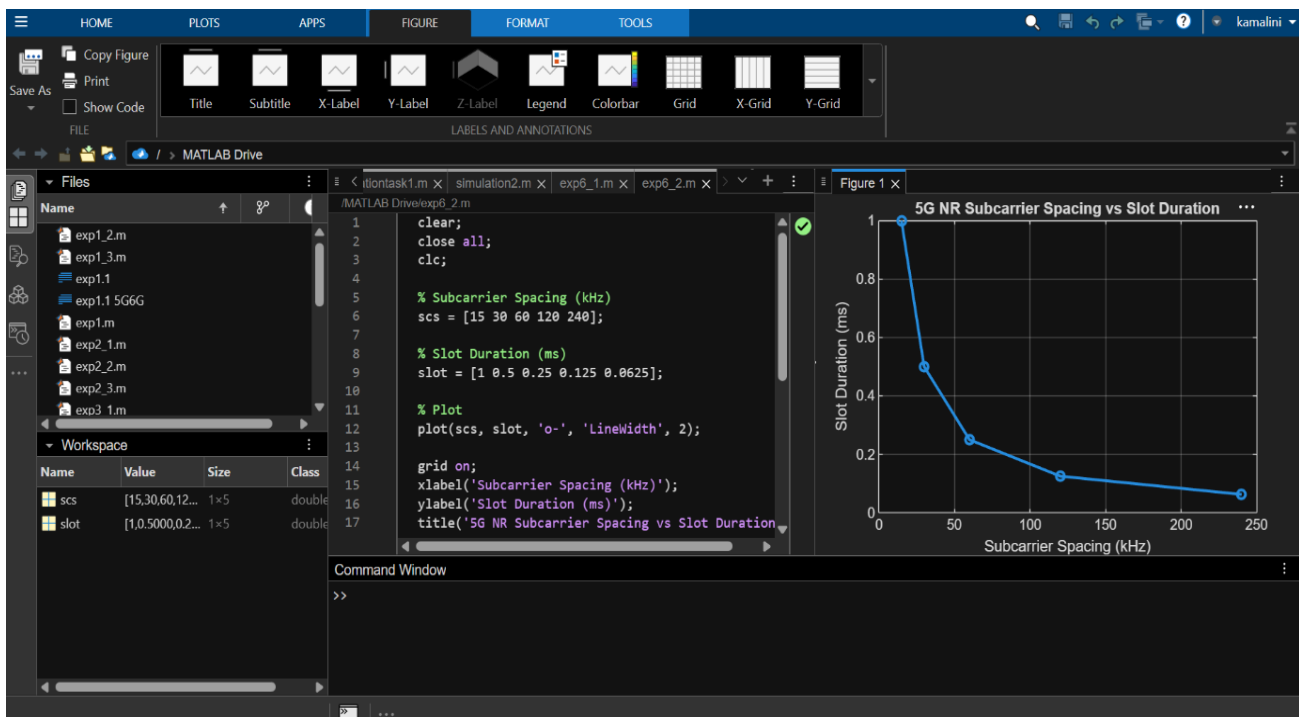
plot(scs, slot, 'o-', 'LineWidth', 2)

grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Slot Duration (ms)')

title('5G NR Slot Duration for Different Subcarrier Spacing')
```



Objectives 3:

To study the number of OFDM symbols present in a 5G NR slot for normal cyclic prefix.

CODE:

```
clc;

clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

symbols = [14 14 14 14 14]; % OFDM Symbols per Slot

figure

bar(scs, symbols)

grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Number of OFDM Symbols')

title('OFDM Symbols per Slot for Different Subcarrier Spacing')

ylim([0 16])
```

